

Probability PhD Exam, May 2003

Write clearly. State precisely results you use. Do not omit steps.

1. Let $(\bar{X}_m)_{m=1}^\infty$ be an i.i.d. sequence of real valued variables on $(\Omega, \mathcal{F}, P)$. Let $\mathcal{F}_m = \sigma(\bar{X}_1, \dots, \bar{X}_m)$. Put $S_m = \bar{X}_1 + \dots + \bar{X}_m$, for $m \geq 1$, and put $S_0 = 0$. Which of the following sequences are martingales relative to $\mathcal{F}_m$? For those which are not martingales with respect to this filtration, what conditions have to be added to make them martingales?
 - a) S_m , $m = 0, 1, 2, \dots$, $E[|\bar{X}_1|] < \infty$.
 - b) $\bar{X}_1^2 + \dots + \bar{X}_m^2 - m\lambda$, $m = 1, 2, \dots$, $E(\bar{X}_1^2) < \infty$, λ real.
 - c) $\exp(\alpha S_m - m\lambda)$, $m = 0, 1, 2, \dots$, where $\varphi(\alpha) = E[\exp(\alpha \bar{X}_1)] < \infty$, α, λ real.
 - d) $\bar{Y}_m = S_{\min(m, T)}$, $m = 0, 1, 2, \dots$, where $P(\bar{X}_1 = \pm 1) = \frac{1}{2}$ and $T = \min\{n > 0 : S_n = 0\}$.
2. Let $\{\alpha_m\}_{m=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$. Show the following are equivalent:
 - a) There exists a probability measure α on $\mathbb{R}$ such that $\lim_{m \rightarrow \infty} \alpha_m(I) = \alpha(I)$ for all closed intervals I on $\mathbb{R}$ whose end points are continuity points of α .
 - b) If $\{F_m\}$, respectively F , are the distribution functions of $\{\alpha_m\}$, respectively α , then $\lim_{m \rightarrow \infty} F_m(x) = F(x)$ at every point x of continuity of F .
3. Let $\{\alpha_m\}_{m=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$, and let $\{\varphi_m(t)\}_{m=1,2,\dots}$ be the corresponding characteristic functions. Assume that $\lim_{m \rightarrow \infty} \varphi_m(t) = \varphi(t)$ for all $t \in \mathbb{R}$, and $\varphi(t)$ is continuous at $t = 0$. Prove that $\varphi(t)$ is the characteristic function of some probability measure and that the conditions of Problem 2 hold.
4. If $\bar{X}$ is a real valued random variable, show that for $0 < \varepsilon < 1$,

$$E \left[\frac{|\bar{X}|}{1 + |\bar{X}|} \right] \leq \varepsilon + P[|\bar{X}| > \varepsilon]$$

and that

$$P(|\bar{X}| > \varepsilon) \leq \frac{1 + \varepsilon}{\varepsilon} E \left[\frac{|\bar{X}|}{1 + |\bar{X}|} \right].$$

Deduce that the set of real valued random variables on a given probability space can (with suitable identification) be regarded as a metric space if the distance between two random variables $\bar{X}$ and $\bar{Y}$ is defined to be

$$d(\bar{X}, \bar{Y}) = E \left[\frac{|\bar{X} - \bar{Y}|}{1 + |\bar{X} - \bar{Y}|} \right],$$

and that convergence in this metric space corresponds to convergence in probability.

5. This problem has three parts.
 - a) Let $\bar{X}_m$, $m = 1, 2, \dots$, be i.i.d., $\bar{X}_m > 0$, $E[\bar{X}_m] = 1$, and $\bar{Y}_m = \prod_{k=1}^m \bar{X}_k$. Prove that $\bar{Y}_m$ is a martingale relative to the filtration $\mathcal{F}_m = \sigma(\bar{X}_1, \dots, \bar{X}_m)$. Show that $\lim_{m \rightarrow \infty} \bar{Y}_m = \bar{Y}_\infty$ exists a.s.
 - b) Show that $P[\bar{Y}_\infty = 0] = 0$ or 1 .
Hint: Recall infinite products and the zero-one law.

c) Using b), show that

$$E[\bar{Y}_\infty] = 1 \iff P[\bar{Y}_\infty > 0] = 1.$$

6. Let $\bar{X}_m$, $m = 1, 2, \dots$, be a supermartingale such that

$$E[|\bar{X}_{m+1} - \bar{X}_m| \mid \mathcal{F}_m] < M \quad \text{a.s.}$$

Show that for any two optional times $S \leq T$, $E[T] < \infty$, we have $E[|\bar{X}_T|] < \infty$ and

$$E(\bar{X}_T \mid \mathcal{F}_S) \leq \bar{X}_S.$$